

Outputs

```
===== MAIN MENU =====
1. Load Dataset
2. Explore Data
3. Perform DataFrame Operations
4. Clean and Handle Missing Data
5. Generate Descriptive Statistics
6. Data Visualization
7. Save Last Plot
8. Exit
Enter your choice: 1
Enter CSV file path: Titanic.csv

File loaded successfully!
```

```
Enter your choice: 2

=== Explore Data ===
1. Display First 5 Rows
2. Display Last 5 Rows
3. Display Column Names
4. Display Data Types
5. Display Basic Info
6. Go Back
Enter your choice: 1
  PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

      Name               Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris   male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0      1
2    Heikkinen, Miss. Laina  female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4    Allen, Mr. William Henry   male  35.0      0

   Parch     Ticket     Fare Cabin Embarked
0      0  A/5 21171   7.2500   NaN        S
1      0    PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0   113803   53.1000  C123        S
4      0   373450   8.0500   NaN        S
```

```
=== Explore Data ===
1. Display First 5 Rows
2. Display Last 5 Rows
3. Display Column Names
4. Display Data Types
5. Display Basic Info
6. Go Back
Enter your choice: 5
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp           891 non-null   int64
7   Parch           891 non-null   int64
8   Ticket          891 non-null   object
9   Fare            891 non-null   float64
10  Cabin           204 non-null   object
11  Embarked        889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
Enter your choice: 3

--- OPERATIONS MENU ---
1. Convert to NumPy
2. Mathematical Operations
3. Combine Datasets
4. Split Dataset
5. Search / Sort / Filter
6. Aggregate Functions
7. Back to Main Menu
Enter operation: 1
Enter column names to convert to NumPy (comma-separated, or leave blank for all):
```

```
Data successfully converted into a NumPy array.
Enter row index to access (leave blank to skip): 0
Enter column index to access (leave blank to skip): 6
Value at row 0, column 6: 1
Enter row slice (start:end, leave blank for all): 0:6
Enter column slice (start:end, leave blank for all): 2:6
```

```
Sliced array:
[[3 'Braund, Mr. Owen Harris' 'male' 22.0]
 [1 'Cumings, Mrs. John Bradley (Florence Briggs Thayer)' 'female' 38.0]
 [3 'Heikkinen, Miss. Laina' 'female' 26.0]
 [1 'Futrelle, Mrs. Jacques Heath (Lily May Peel)' 'female' 35.0]
 [3 'Allen, Mr. William Henry' 'male' 35.0]
 [3 'Moran, Mr. James' 'male' nan]]
```

```
Enter operation: 2
```

```
Numeric columns available: ['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']
```

```
Available operations:
```

1. Add two columns
2. Subtract two columns
3. Multiply two columns
4. Divide two columns
5. Percentage of one column over another
6. Compute mean or sum of selected columns
7. Back to main menu

```
Enter operation number: 1
Enter first numeric column: SibSp
Enter second numeric column: Parch
Enter new column name for result: Fmembers
```

```
Fmembers = SibSp + Parch
```

```
First 5 rows after operation:
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked	Fmembers
0	0	A/5 21171	7.2500	NaN	S	1
1	0	PC 17599	71.2833	C85	C	1
2	0	STON/O2. 3101282	7.9250	NaN	S	0
3	0	113803	53.1000	C123	S	1
4	0	373450	8.0500	NaN	S	0

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Enter your choice: 4
```

```
=== Data Cleaning Started ===
Filled missing numeric values in 'Age' with mean (29.70).
Filled missing categorical values in 'Cabin' with 'Unknown'.
Filled missing categorical values in 'Embarked' with 'Unknown'.

Cleaning Completed:
- Rows before: 891
- Rows after: 891
- Missing values before: 866
- Missing values after: 0
=====
```

```
Enter your choice: 5

--- STATISTICS MENU ---
1. Statistical Analysis
2. Pivot Table
3. Back to Main Menu
Enter choice: 1
Enter column name for statistical analysis: Fare
```

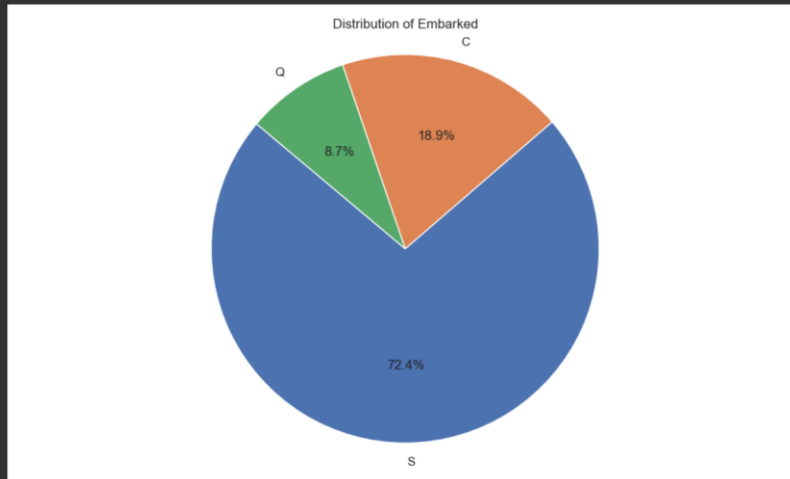
```
Statistical Analysis for 'Fare':
Standard Deviation: 49.69
Variance: 2469.44
Quantiles:
0.25    7.9104
0.50   14.4542
0.75   31.0000
Name: Fare, dtype: float64
```

```
--- STATISTICS MENU ---
1. Statistical Analysis
2. Pivot Table
3. Back to Main Menu
Enter choice: 2
Enter index columns (comma-separated): Embarked
Enter value columns (comma-separated): Sex
Enter aggregation function (sum, mean, count): count

Pivot Table:
          Sex
Embarked
C         168
Q          77
S        644
```

```
==== Visualization Menu ====
1. Bar Plot
2. Box Plot
3. Scatter Plot
4. Histogram
5. Heatmap
6. Pie Chart
7. Stack Plot
8. Go Back
Enter plot type number: 6
Enter column for pie chart: Embarked

Plot generated. Use option 7 in main menu to save it.
```



```
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Enter your choice: 7

Enter filename to save plot (e.g., plot.png): Embarked.png
Plot saved successfully as 'Embarked.png'.
Figure closed and memory released.

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Enter your choice: 8
Exiting the program. Goodbye!

Analyzer object destroyed. Cleanup completed.
```